

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intelligence Application Development and Jira

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TASK 2

Q.1 In a college:

- 20% of students know Python.
- 80% of students know Java.
- 80% of students know Java.
- 90% of Python students get placed.
- 40% of Java students get placed.

Find:

- Probability that the student knows Python.
- Probability that the student knows Java.
- Probability that a placed student knows Python.

CODE :-

```
# Probability of knowing Python

P_Python = 0.20

# Probability of knowing Java

P_Java = 0.80

# Probability of getting placed if student knows Python

P_Placed_Python = 0.90

# Probability of getting placed if student knows Java

P_Placed_Java = 0.40

# Probability that a placed student knows Python (Bayes Rule)

P_Python_Placed = (P_Placed_Python * P_Python) / ((P_Placed_Python *
P_Python) + (P_Placed_Java * P_Java))
```

```

print("Probability that the student knows Python :")

print(P_Python)

print("Probability that the student knows Java :")

print(P_Java)

print("Probability that a placed student knows Python :")

print(round(P_Python_Placed * 100, 2), "%")

print("Saqlain Khan T013")

```

OUTPUT :-

```

Probability that the student knows Python :
0.2
Probability that the student knows Java :
0.8
Probability that a placed student knows Python :
36.0 %
Saqlain Khan T013

```

Q.2 In a city:

- 85% of taxis are Green.
- 15% are Blue.
- A witness correctly identifies the taxi color 80% of the time.

A witness claims the taxi involved in an accident was Blue. What is the probability that the taxi was actually Blue?

CODE :-

```

# Probability of Blue taxi

P_Blue = 0.15

# Probability of Green taxi

P_Green = 0.85

# Probability that witness says Blue if taxi is actually Blue

P_SaysBlue_Blue = 0.80

```

```
# Probability that witness says Blue if taxi is actually Green

P_SaysBlue_Green = 0.20

# Bayes Rule

P_Blue_SaysBlue = (P_SaysBlue_Blue * P_Blue) / ((P_SaysBlue_Blue *
P_Blue) + (P_SaysBlue_Green * P_Green))

print("Probability that taxi is actually Blue :")

print(round(P_Blue_SaysBlue * 100, 2), "%")

print("Saqlain Khan T013")
```

OUTPUT :-

```
Probability that taxi is actually Blue :
41.38 %
Saqlain Khan T013
```